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(54) **LIGHTING DEVICE FOR BEVERAGE CONTAINER, LIGHTING SYSTEM FOR BEVERAGE CONTAINER AND METHOD OF LIGHTING IN BEVERAGE CONTAINER**

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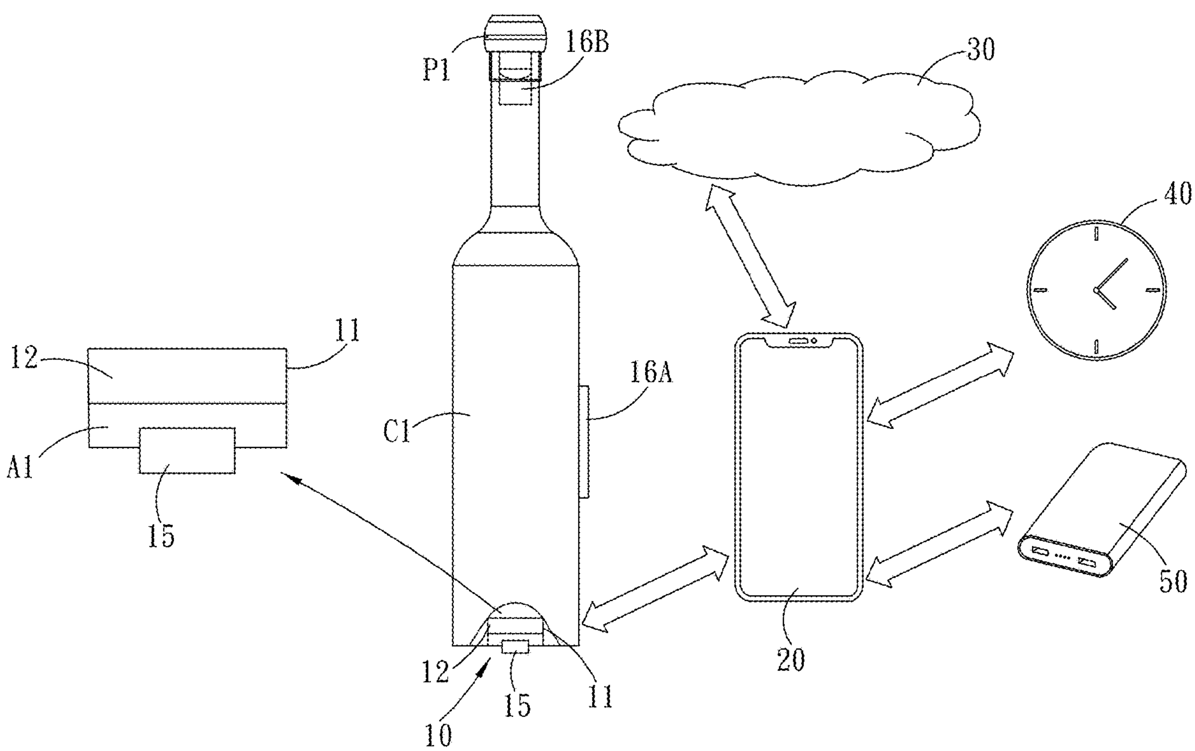
(57) **ABSTRACT**

A lighting device for a beverage container includes a base, at least one light source, a wireless transceiver, and a microprocessor. The base is affixed to the beverage container. At least one light source is disposed on the base and is configured to emit light. The wireless transceiver is disposed within the base and is configured to receive audio information and sensing information. The microprocessor is disposed within the base, is electrically connected to at least one light source and the wireless transceiver, and controls at least one light source to change at least one of the colors, the light intensity and the operation frequency of the light according to the audio information and the sensing information. Based on the aforementioned configuration, the pattern of the light varies within the beverage container according to the audio information and the sensing information, thereby creating vibrant visual experience.



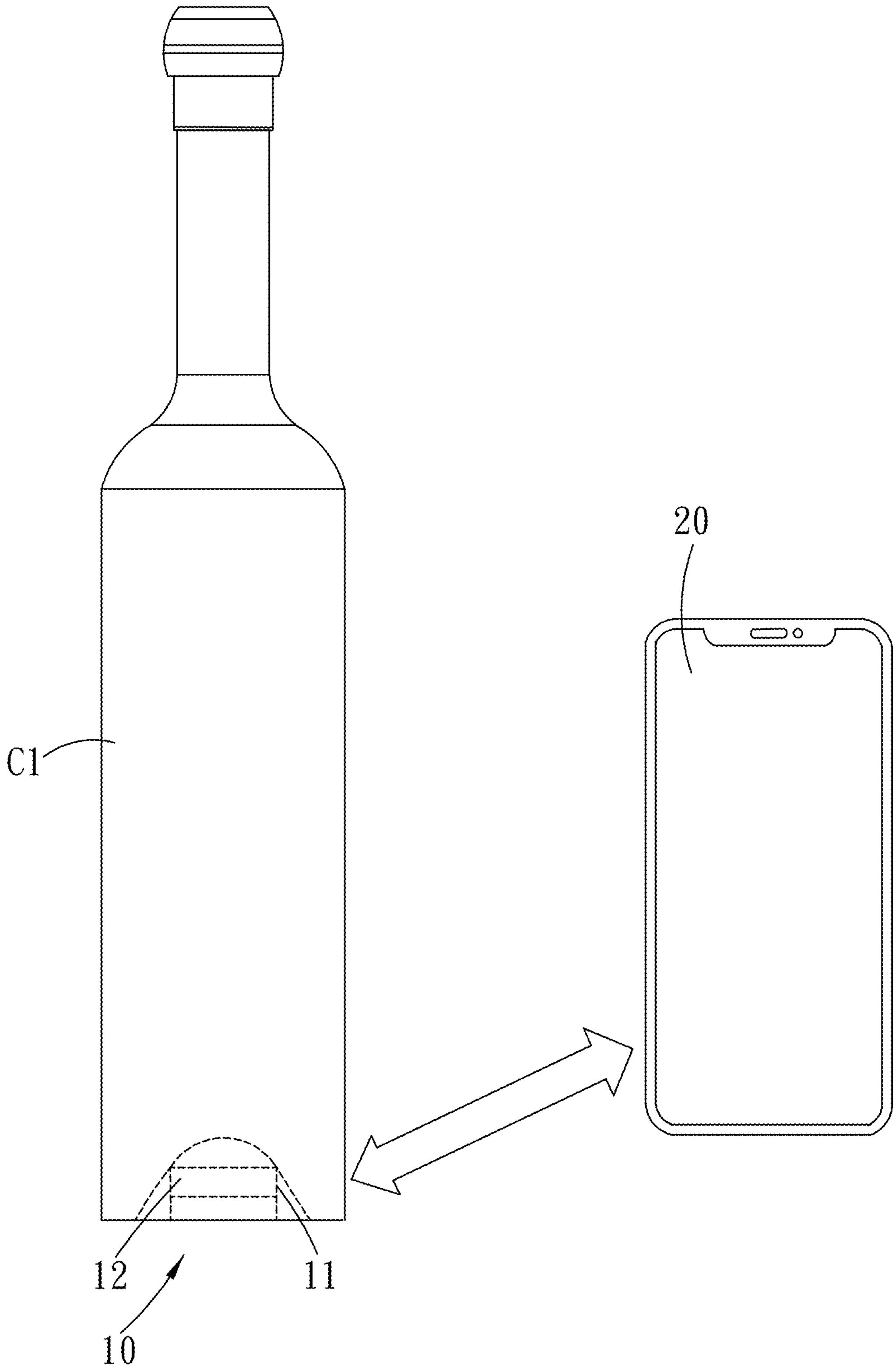


FIG. 1A

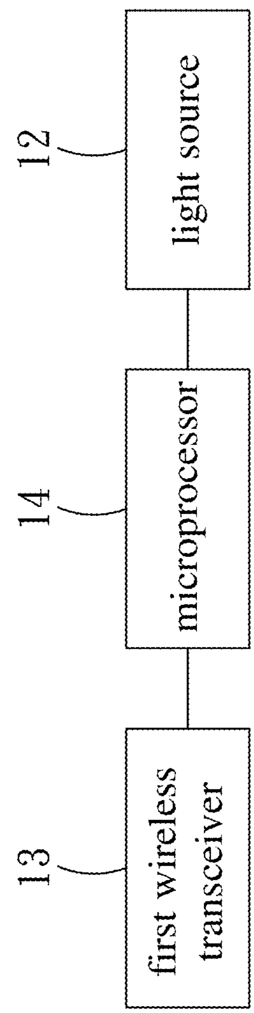

12
light source
14
microprocessor
13
first wireless transceiver

FIG. 1B

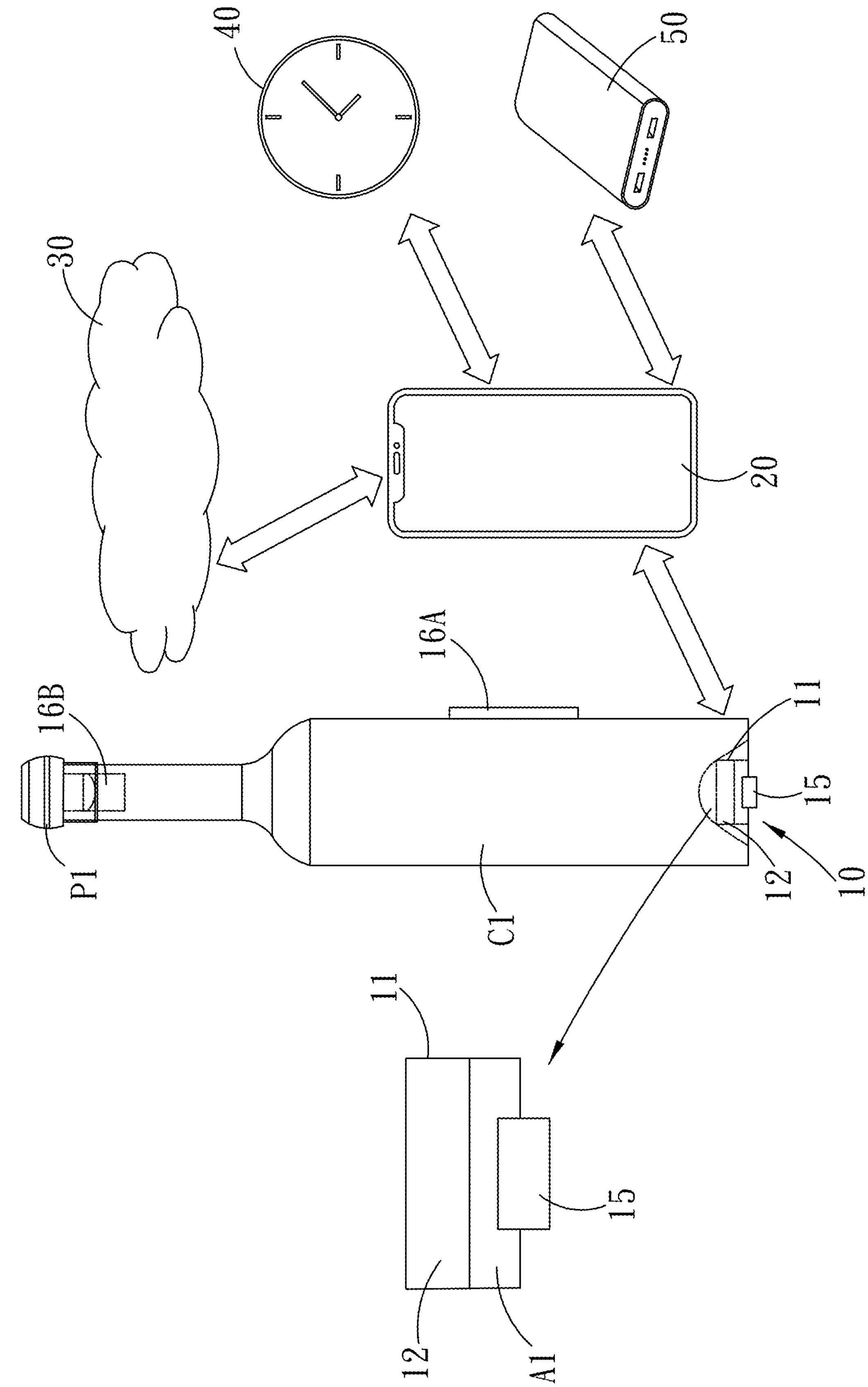


FIG. 2A

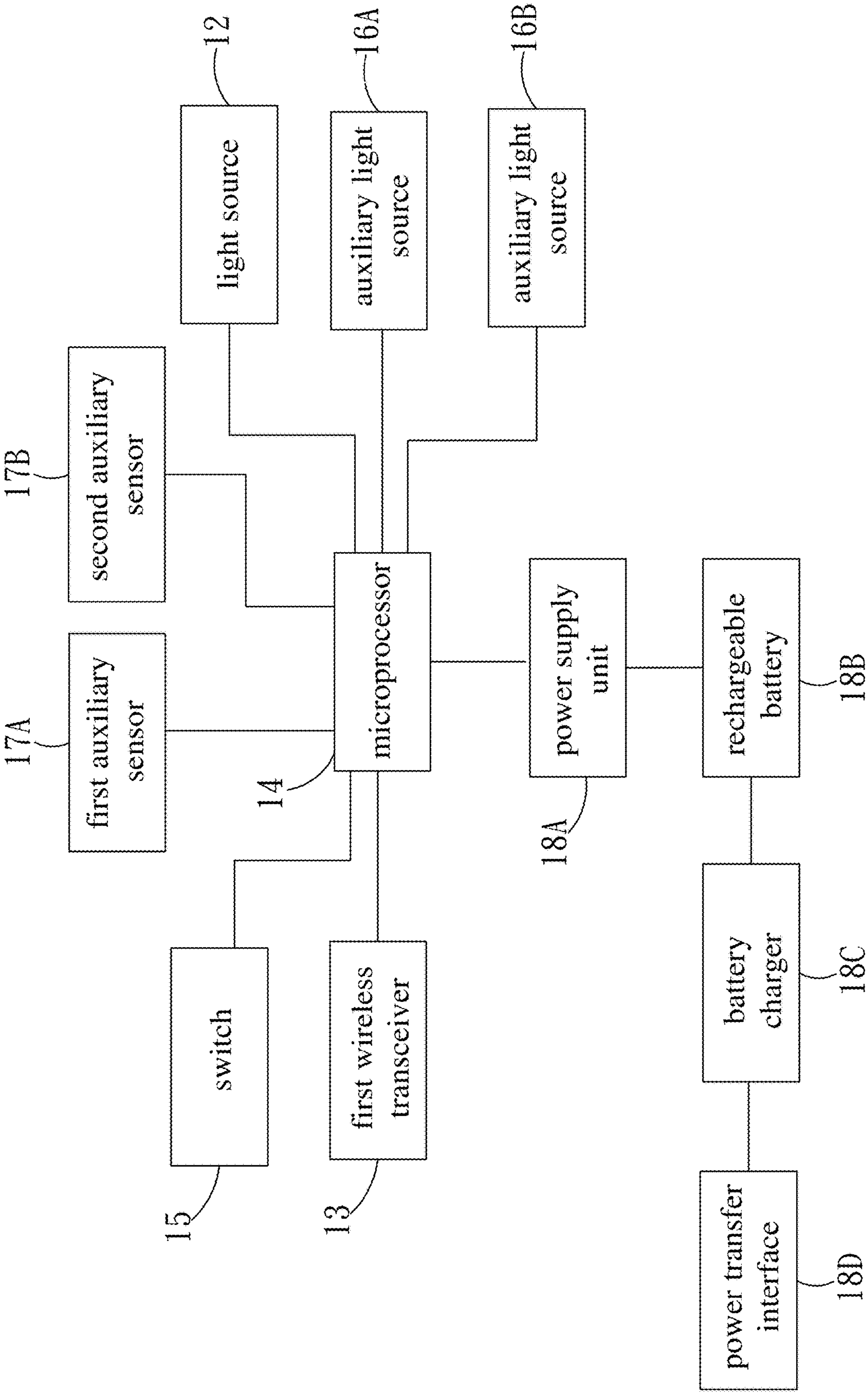
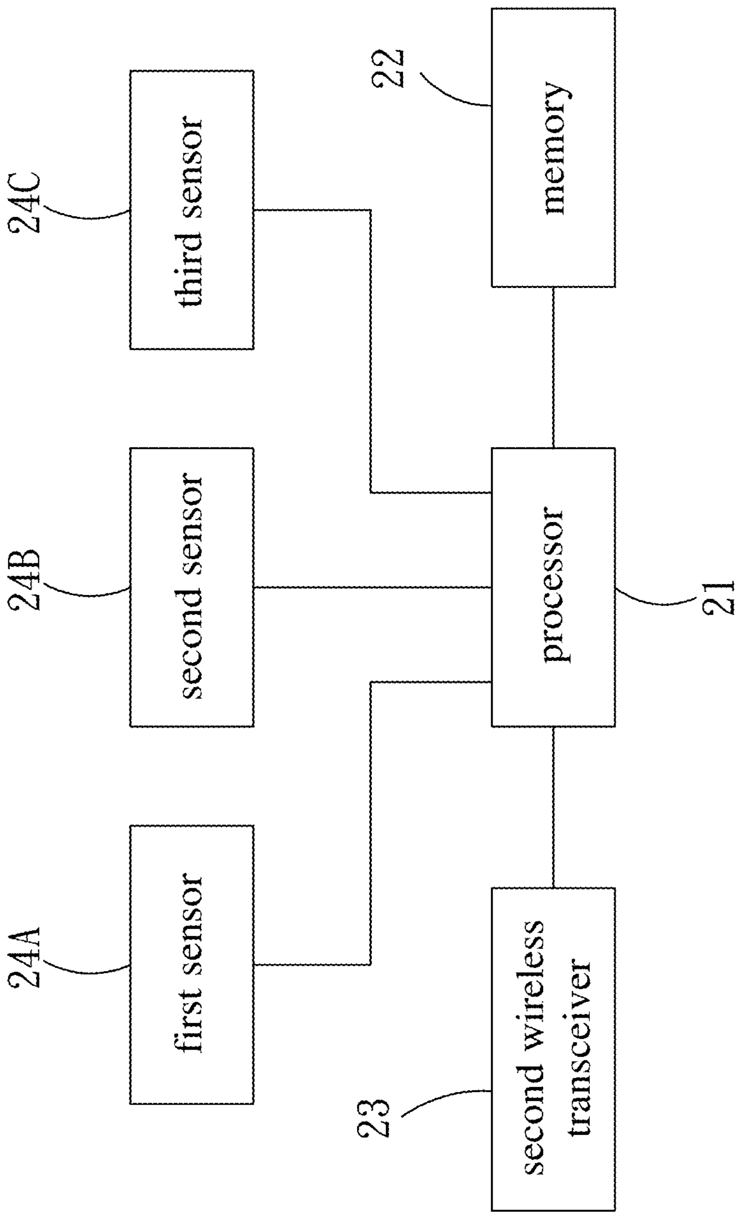

12
light source
16A
auxiliary light source
16B
auxiliary light source
17B
second auxiliary sensor
17A
first auxiliary sensor
14
microprocessor
power supply unit
18A
rechargeable battery
18B
battery charger
18C
power transfer interface
18D
switch
15
first wireless transceiver
13

FIG. 2B



third sensor
second sensor
first sensor
memory
processor
second wireless transceiver
24C
24B
24A
22
21
23

FIG. 2C

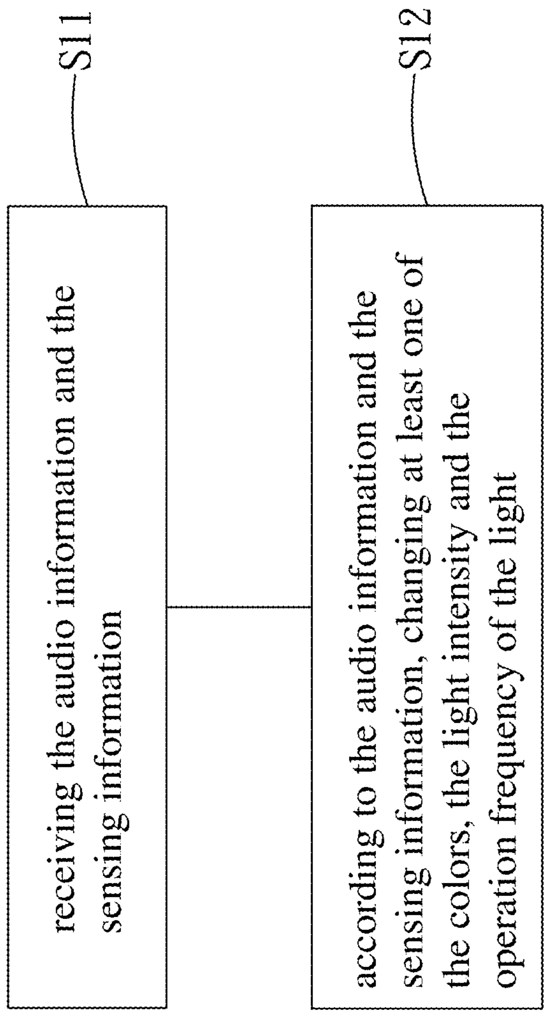

S11
receiving the audio information and the sensing information
S12
according to the audio information and the sensing information, changing at least one of the colors, the light intensity and the operation frequency of the light

FIG. 3

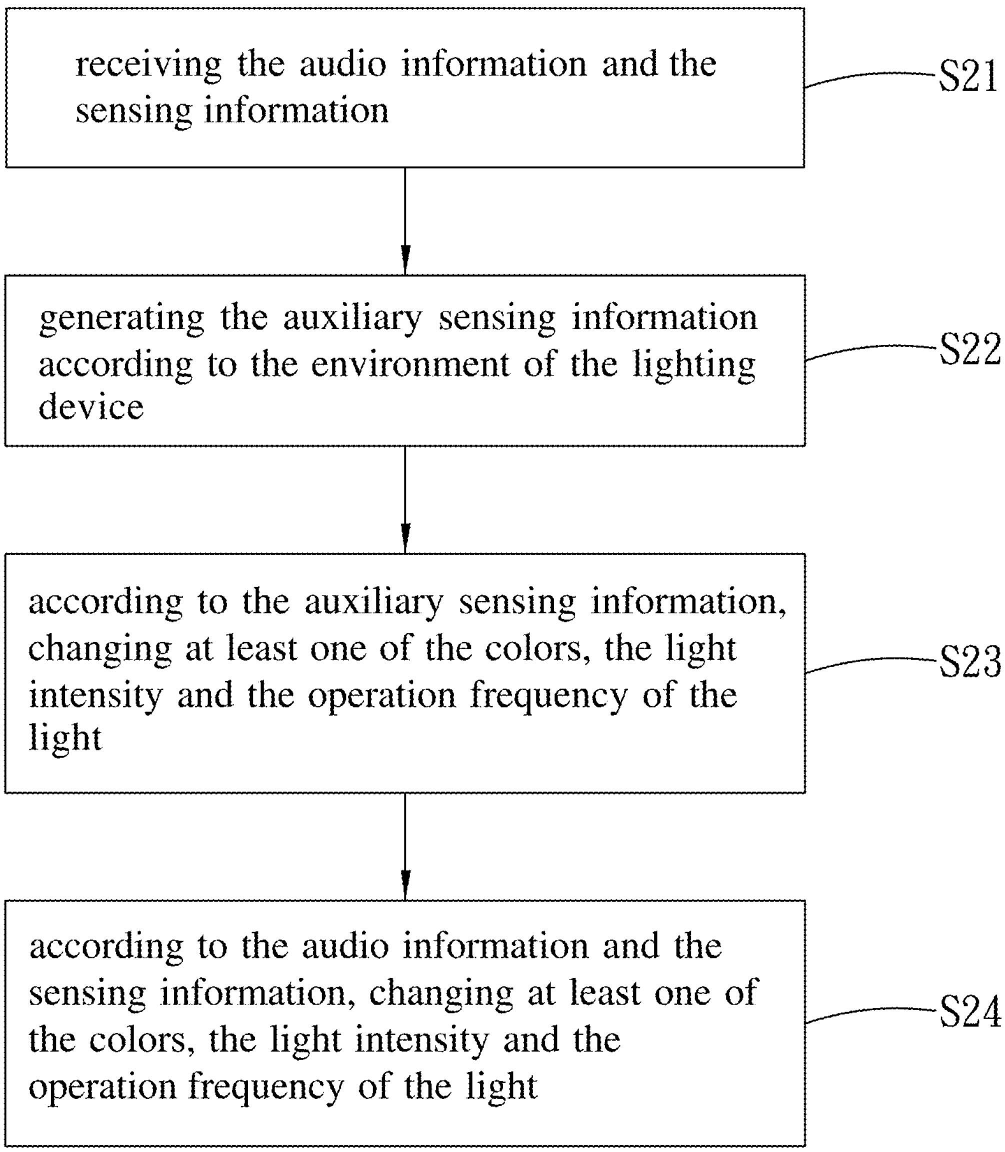

receiving the audio information and the sensing information
S21
generating the auxiliary sensing information according to the environment of the lighting device
S22
according to the auxiliary sensing information, changing at least one of the colors, the light intensity and the operation frequency of the light
S23
according to the audio information and the sensing information, changing at least one of the colors, the light intensity and the operation frequency of the light
S24

FIG. 4

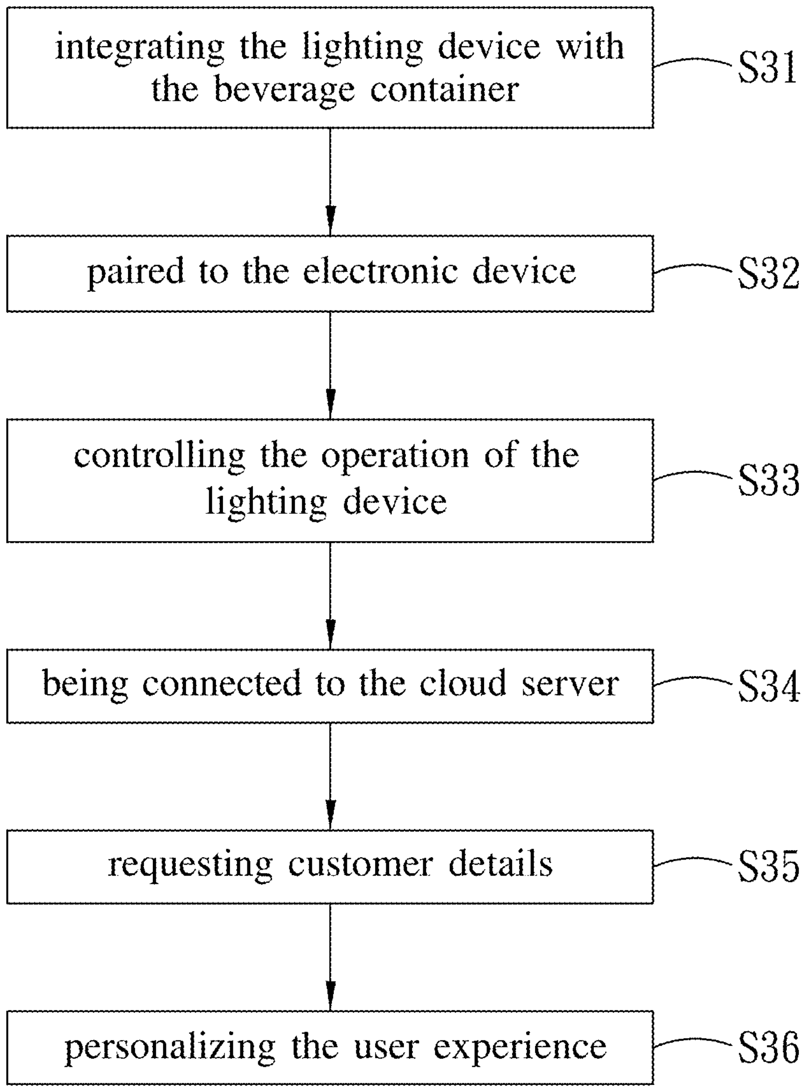

integrating the lighting device with the beverage container
S31
paired to the electronic device
S32
controlling the operation of the lighting device
S33
being connected to the cloud server
S34
requesting customer details
S35
personalizing the user experience
S36

FIG. 5

LIGHTING DEVICE FOR BEVERAGE CONTAINER, LIGHTING SYSTEM FOR BEVERAGE CONTAINER AND METHOD OF LIGHTING IN BEVERAGE CONTAINER

TECHNICAL FIELD OF THE INVENTION

[0001] This non-provisional application claims priority claim under 35 U.S.C. § 119 (a) on U.S. Provisional Patent Application No. 63/424,272 filed Nov. 10, 2022 the entire contents of which are incorporated herein by reference. The present disclosure is related to the technical field of the dynamic landscape of beverage containers in consumer markets and is particularly directed to a lighting device for a beverage container, a lighting system for a beverage container and a method of lighting in a beverage container.

DESCRIPTION OF THE PRIOR ART

[0002] The current trend of beverage containers in consumer markets focuses on the various applications of wireless technologies such as Bluetooth in beverage containers. For example, U.S. Pat. No. 9,004,320B2 describes a mechanism to control the flow of a beverage from a bottle by the wireless communication between a pour spout and a control unit, and U.S. Pat. No. 11,237,036B2 describes a mechanism to monitor the content of a container by a sensor device.

[0003] However, there is no development or product about the beverage container to create a direct consumer connection which fosters the real-time interaction between a consumer and the beverage container. How to create the direct consumer connection in the beverage container becomes an important issue.

SUMMARY

[0004] In light of the aforementioned descriptions, the present disclosure provides a lighting device for a beverage container, a lighting system for a beverage container and a method of lighting in a beverage container to solve the problem in creating the direct consumer connection in the beverage container.

[0005] Based on the aforementioned descriptions, the present disclosure provides the lighting device for the beverage container. The lighting device for the beverage container includes a base, at least one light source, a wireless transceiver and a microprocessor. The base is affixed to the beverage container. At least one light source is disposed on the base and is configured to emit light. The wireless transceiver is disposed within the base and is configured to receive audio information and sensing information. The microprocessor is disposed within the base, is electrically connected to at least one light source and the wireless transceiver, and controls at least one light source to change at least one of the colors, the light intensity and the operation frequency of the light according to the audio information and the sensing information.

[0006] Based on the aforementioned descriptions, the present disclosure provides the lighting system for the beverage container. The lighting system for the beverage container includes an electronic device and the lighting device. The electronic device transmits audio information and sensing information. The lighting device communicates with the electronic device and includes a base, at least one light source, a wireless transceiver and a microprocessor. The base is affixed to the beverage container. At least one light source is disposed on the base and is configured to emit light. The wireless transceiver is disposed within the base and is configured to receive audio information and sensing information. The microprocessor is disposed within the base, is electrically connected to at least one light source and the wireless transceiver, and controls at least one light source to change at least one of the colors, the light intensity and the operation frequency of the light according to the audio information and the sensing information.

[0007] Based on the aforementioned descriptions, the present disclosure provides the method of lighting in the beverage container. The method of lighting in the beverage container is for a lighting system, and the lighting system includes an electronic device and a lighting device, and the lighting device is configured to emit light and is affixed to the beverage container. The method of lighting in the beverage container includes performing steps as follows by the lighting device: receiving audio information and sensing information from the electronic device; according to the audio information and sensing information, changing at least one of the colors, the light intensity and the operation frequency of the light.

[0008] In view of the above description, in the lighting device for the beverage container, the lighting device and the beverage container integrate seamlessly, and the pattern of the light varies within the beverage container according to the audio information and the sensing information, thereby creating vibrant visual experience, thus enhancing the user experience of the beverage container with the lighting device and encouraging interactive communication with customers.

[0009] In view of the above description, in the lighting system for the beverage container and the method of lighting in the beverage container, the lighting device receives plenty of real-time data such as the audio information and the sensing information from the electronic device to synchronize the pattern of the light with the music in the audio information and to transform the sensing information into the dynamic pattern of the light, thereby fostering an enriching experience with the beverage container having the lighting device.

[0010] In addition, when the beverage container with the lighting device is exhibited, the beverage container with the lighting device not only engage consumers at a deeper level but also transforms the consumer behavior of purchasing and experiencing beverage in the beverage container into a more immersive and enjoyable journey.

[0011] The aforementioned description of the present disclosure is merely the outline of the technical solutions of the present disclosure. In order to understand the technical solutions of the present disclosure clearly and to implement the present disclosure according to the content of the specification, the better embodiments of the present disclosure given herein below with accompanying drawings are used to describe the present disclosure in detail.

BRIEF DESCRIPTION OF DRAWINGS

[0012] FIG. **1**A depicts a schematic diagram of a lighting system for a beverage container according to one embodiment of the present disclosure.

[0013] FIG. **1**B depicts the block diagram of the lighting device for the beverage container shown in FIG. **1**A.

[0014] FIG. **2**A depicts a schematic diagram of a lighting system for a beverage container according to another embodiment of the present disclosure.

[0015] FIG. **2**B depicts the block diagram of the lighting device for the beverage container shown in FIG. **2**A.

[0016] FIG. **2**C depicts the block diagram of the electronic device shown in FIG. **2**A.

[0017] FIG. **3** depicts a flowchart of a method of lighting in a beverage container according to one embodiment.

[0018] FIG. **4** depicts a flowchart of a method of lighting in a beverage container according to another embodiment.

[0019] FIG. **5** depicts a flowchart of a method of lighting in a beverage container according to yet embodiment.

DETAILED DESCRIPTION

[0020] The specific embodiments of the present disclosure given herein below is used to explain the implementation of the present disclosure. A person skilled in the art easily understands the advantages and the effects of the present disclosure from the content of the present disclosure.

[0021] It should be noted that the embodiments and the features in the embodiments of the present disclosure can be combined with each other without conflict. The present disclosure will be described in detail below with reference to accompanying drawings and in conjunction with the embodiments. In order to provide those in the art with better understanding of the solution of the disclosure, the technical solutions in the embodiments of the present disclosure will be described clearly and completely below in conjunction with the accompanying drawings in the embodiments of the present disclosure. Apparently, the described embodiments are merely one part of the embodiments of the present disclosure and not all embodiments of the present disclosure. Based on the embodiments of the present disclosure, all embodiments obtained by a person skilled in the art without any inventive steps shall fall within the scope of protection of the present disclosure.

[0022] It should be noted that the terms "first", "second", etc. in the specification and claims of the present disclosure and in the aforementioned accompanying drawings are used to distinguish similar objects and not used to describe a particular order or sequence. Furthermore, the terms "comprising" and "having", and any variation thereof, are intended to encompass a non-exclusive inclusion, for example, a series of steps or units comprising processes, methods, systems, products or equipment do not need to be limited to those steps or units clearly listed but may include other steps or units not clearly listed or inherent to those processes, methods, products or equipment.

[0023] Please refer to FIG. **1**A, which depicts a schematic diagram of a lighting system for a beverage container according to one embodiment of the present disclosure. As shown in FIG. **1**A, a lighting system for a beverage container C**1** includes a lighting device **10** and an electronic device **20**. The beverage container C**1** accommodates a beverage such as milk, coffee and whisky and may be a bottle or a glass container.

[0024] The electronic device **20** is wirelessly connected to the lighting device **10** to communicate with the lighting device **10** and may be a mobile phone, a computer or a smart speaker. The wireless connection between the electronic device **20** and the lighting device **10** may be but not be limited to a Bluetooth connection, a Wi-Fi connection, or a ZigBee connection. The electronic device **20** transmits plenty of data to the lighting device **10** by the wireless connection to build a data-rich environment for the lighting device **10**, and the plenty of data include audio information, sensing information and the other types of information.

[0025] The lighting device **10** is embedded within the beverage container C**1** for ornamental lighting and is located on the bottom of the beverage container C**1**, for example, but the position of the lighting device **10** may be changed to the sidewall of the beverage container C**1** and not be limited thereto. The light device **10** is configured to generate light to illuminate the interior of the beverage container C**1** and varies the pattern and the mode of the light according to the audio information and the sensing information to provide an interactive user experience and to enhance the visual experience of the light. Hence, the pattern and the mode of the light are influenced by the audio information and the sensing information. In other words, the electronic device **20** indirectly controls the pattern and the mode of the light and is regarded as the controller for the pattern and the mode of the light.

[0026] Please refer to FIG. **1**B, which depicts the block diagram of the lighting device for the beverage container shown in FIG. **1**A. As shown in FIG. **1**B, in conjunction with FIG. **1**A, the lighting device **10** includes a base **11**, at least one light source **12**, a first wireless transceiver **13** and a microprocessor **14**.

[0027] The base **11** is affixed to the beverage container C**1**. Specifically, the base **11** is fixed at the bottom of the beverage container C**1** by snap-fit structures and is located at a recess on the bottom of the beverage container C**1**. The interior of the base **11** accommodates at least one light source **12**, the first wireless transceiver **13** and the microprocessor **14**. The bottle preferably has special design printed on the glass that makes the light from the base **11** illuminate all its ornamentals and create elevating lightning show effects.

[0028] At least one light source **12** is disposed on the base **11**, is electrically connected to the microprocessor **14**, and is configured to emit the light. In present embodiment, at least one light source **12** is disposed within the base **11** and is located on the interior of the base **11**. In another embodiment, at least one light source **12** protrudes from the top surface of the base **11** and contacts the recess on the bottom of the beverage container C**1**. At least one light source **12** is controlled by the microprocessor **14** to selectively emit light, and the color, the light intensity, and the operation frequency of the light is determined by the microprocessor **14**. At least one light source **12** may be one or more light-emitting diodes (LEDs), and the color of at least one light source **12** may be single or multiple, thereby providing versatile illumination options. In the present embodiment, the number of at least one light source **12** is one, and the color of the light source **12** is single or multiple. In another embodiment, the number of at least one light source **12** is plural, and the color of each light source **12** is single.

[0029] The first wireless transceiver **13** is disposed within the base **11**; in other words, the first wireless transceiver **13** is located on the interior of the base **11**. The first wireless transceiver **13** is electrically connected to the microprocessor **14**, and is configured to receive the plenty of data including the audio information and the sensing information from the electronic device **20**. Specifically, the first wireless transceiver **13** serves as the IoT (Internet of Things) interface of the lighting device **10** to enable the remote control and the monitoring of the electronic device **20**. The communication protocol of the first wireless transceiver **13** may be but not be limited to Bluetooth or Wi-Fi.

[0030] The microprocessor **14** is disposed within the base; in other words, the microprocessor **14** is located on the interior of the base **11**. The microprocessor **14** controls the operation of the light source **12** and the operation of the first wireless transceiver **13**, and controls the light source **12** to change at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and the sensing information. The details of controlling the light source **12** to change at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and the sensing information would be elaborated in the paragraphs of a method of lighting in a beverage container.

[0031] In the present embodiment, there is a circuit board on the interior of the base **11**, and the circuit board may be but not be limited to a printed circuit board (PCB) or a flexible printed circuit board (FPCB). The first wireless transceiver **13** and the microprocessor **14** are disposed on the circuit board, and the light source **12** and the first wireless transceiver **13** are electrically connected to the microprocessor **14** by conductive wires.

[0032] Please refer to FIG. **2**A, which depicts a schematic diagram of a lighting system for a beverage container according to another embodiment of the present disclosure. As shown in FIG. **2**A, a lighting system for a beverage container C**1** includes the lighting device **10** and the electronic device **20**. The similarities between the configuration of FIG. **1**A and the configuration of FIG. **2**A would not be repeated.

[0033] The electronic device **20** is connected to a cloud server **30** and a media input source **40** by Internet, is connected to an external storage device **50** by a universal serial bus (USB) connector and a USB cable, and obtains environment information and the audio information from the cloud server **30**, the media input source **40** and the external storage device **50**. In other words, the electronic device **20** serves as an IoT-enabled device.

[0034] The environment information may include weather information, time information and geographical information which are changed according to the geographical position of the electronic device **20**. The weather information includes the different weather of the different geographical positions, and the time information includes the different time of the different geographical positions. The geographical information includes the different local cultures at the different geographical positions, the different ambiances of specific neighborhoods or street clubs at the different geographical positions, and the different local festivals at the different geographical positions. The audio information includes user music, popular music, local music and the other audio files. It should be noted that the user music may be previously stored in the electronic device **20** or be obtained from the external storage device **50**.

[0035] In addition, the electronic device **20** is provided with application software. The application software establishes a straightforward connection to the cloud server **30**, the media input source **40**, and the external storage device **50** and gathers user information and the sensing information in the electronic device **20** to assist a user in controlling the lighting of the lighting device **10**. In other words, the application software empowers the user to control the lighting of the lighting device **10**. For example, the electronic device **20** is a smart phone, and the application software is a phone application. The phone application assists the user in collecting the user information, the sensing information, and the audio information and remotely controlling the operation of the lighting device **10**.

[0036] The cloud server **30** is a cloud platform which grants the electronic device **20** access to various internet radio stations spanning diverse music styles or cloud music service, and the electronic device **20** obtains the popular music, the local music, and the other audio files from the cloud server **30**. The popular music, the local music and the other audio files constitute one part of the audio information. In addition, the cloud server **30** stores various light templates and the geographical information, and the electronic device **20** is able to capture the needed light template from the cloud server **30** according to user requirements and to select the corresponding local culture and the corresponding ambiance of specific neighborhoods or street clubs from the cloud server **30** according to the geographical position of the electronic device **20** to assist the user in the customization of the lighting effects of the lighting device **10**.

[0037] The media input source **40** is an information station to provide the electronic device **20** with the weather information and the time information to assist the user in selecting the appropriate music from the audio information. In addition, the electronic device **20** is able to capture the local news of the geographical position of the electronic device **20** from the media input source **40** to assist the user in confirming what event occurs. The media input source **40** may be a home weather station, a smart connected clock station, or a television (TV) station, and the number of the media input source **40** may be one or more.

[0038] The external storage device **50** provides the electronic device **20** with the user music to assist the user in the personalization of the lighting effects of the lighting device **10**. In addition, the electronic device **20** is able to capture a user video from the external storage device **50** to assist the user in the personalization of the lighting effects of the lighting device **10**. The user music and the user video constitute the other part of the audio information. The external storage device **50** is a portable storage device or a USB flash drive.

[0039] In addition, the user uses the application software in the electronic device **20** to automatically gather or collect the user information, the sensing information, the audio information and the environment information, and the application software provides the appropriate light templates for the user according to the user information, the sensing information, the audio information, and the environment information. When the lighting device **10** is provided in plural, the user uses the application software to indirectly control the operation of the lighting devices **10** or to control the lighting devices **10** to operate according to the light template which the user selects.

[0040] According to the above descriptions, the electronic device **20** can retrieve and play any available music files stored in the cloud server **30** and the external storage device **50** according to the user requirements, thereby broadening the listening options of the user and catering to user preferences. In addition, the application software may integrate a wealth of music according to the user requirements to facilitate the customization of the lighting effects of the lighting device **10**.

[0041] Please refer to FIG. **2**B, which depicts the block diagram of the lighting device for the beverage container shown in FIG. **2**A. As shown in FIG. **2**B, in conjunction with FIG. **2**A, the lighting device **10** includes a base **11**, at least one light source **12**, the first wireless transceiver **13**, the microprocessor **14**, a switch **15**, auxiliary light sources **16**A and **16**B, at least one auxiliary sensor, a power supply unit **18**A, a rechargeable battery **18**B, a battery charger **18**C, a power transfer interface **18**D, and an attaching surface A**1**. The configurations of the base **11**, at least one light source **12**, the first wireless transceiver **13** and the microprocessor **14** shown in FIG. **2**A and FIG. **2**B is similar to the configurations of the base **11**, at least one light source **12**, the first wireless transceiver **13** and the microprocessor **14** shown in FIG. **1**A and FIG. **1**B and would not be repeated. In the present embodiment, the number of at least one auxiliary sensor is two, two auxiliary sensors are a first auxiliary sensor **17**A and a second auxiliary sensor **17**B. However, the number of at least one auxiliary sensor may be adjusted according to the requirements of the lighting device **10** and may be one, three or more than three, and the present disclosure is not limited thereto.

[0042] The attaching surface A**1** is disposed under the bottom surface of the base **11** to be attached to the beverage container C**1**, and the bottom surface of the beverage container C**1** contacts the attaching surface A**1**, thus allowing for the easy attachment of the base **11** to the surfaces of the various bottles or the various glass containers. The materials of the attaching surface A**1** may include but not be limited to friction materials or adhesive materials, and the adhesive materials may include a hot-melt adhesive or a UV adhesive, for example.

[0043] The switch **15** is disposed adjacent to the base **11** and is electrically connected to the microprocessor **14**. Specifically, the switch **15** is disposed under the bottom surface of the base **11**, one part of the switch **15** contact the attaching surface A**1**, and the other part of the switch **15** protrudes from the bottom of the attaching surface A**1** and is exposed to air. There are the conductive wires which start from the microprocessor **14** and penetrate the attaching surface A**1** to contact the switch **15**, and the conductive wires penetrating the attaching surface A**1** connect the switch **15** and the microprocessor **14**.

[0044] The switch **15** is configured to generate a trigger signal. In the present embodiment, the switch **15** is provided with an on button and an off button. When the on button is pressed, the switch **15** generates and transmits the trigger signal to the microprocessor **14**, and the microprocessor **14** controls at least one light source **12** to be turned on. When the off button is pressed, the switch **15** does not generate the trigger signal, and the microprocessor **14** controls at least one light source **12** to be turned off. In other words, the microprocessor **14** controls at least one light source **12** to be turned on or be turned off according to the trigger signal. In another embodiment, the switch **15** is a multi-step switch. The switch **15** is configured to generate and transmit the different trigger signals to the microprocessor **14** according to different touch actions, and the microprocessor **14** generates and transmits different control signals to at least one light source **12**. Afterwards, at least one light source **12** generates the different patterns of the light according to the different control signals, and the different patterns of the light may include static illumination, strobe effects, color transitions, etc.

[0045] The auxiliary light source **16**A is disposed on the sidewall of the beverage container C**1**, while the auxiliary light source **16**B is disposed adjacent to the cap P**1** of the beverage container C**1**. It should be noted that the positions of the auxiliary light sources **16**A and **16**B in FIG. **2**A are exemplary and are not limited thereto. The auxiliary light sources **16**A and **16**B are electrically connected to the microprocessor **14** and are controlled by the microprocessor **14**. The light emitted from the light source **12** is main light, while the light emitted from the auxiliary light sources **16**A and **16**B is auxiliary light. The auxiliary light sources **16**A and **16**B assist the light source **12** in the different patterns of the light. In addition, the auxiliary light sources **16**A and **16**B may have their own switches to facilitate the control of the auxiliary light. In collaboration with the light source **12** and the auxiliary light sources **16**A and **16**B, the colors of the light may encompass a palette of colors.

[0046] The first auxiliary sensor **17**A is located on the interior of the base **11** and is electrically connected to the microprocessor **14**. The microprocessor **14** controls the operation of the first auxiliary sensor **17**A, and when the first auxiliary sensor **17**A generates first auxiliary sensing information according to the environment of the lighting device **10**, the microprocessor **14** receives the first auxiliary sensing information from the first auxiliary sensor **17**A and controls the light source **12** to change at least one of the color, the light intensity or the operation frequency of the light according to the first auxiliary sensing information. For example, the first auxiliary sensor **17**A is a microphone, and the first auxiliary sensing information includes the frequency and the sound level (e.g., sound pressure level, SPL) of ambient sound around the environment of the lighting device **10**. The microprocessor **14** controls the light source **12** to change the color and the light intensity of the light according to the frequency and the sound level of ambient sound to realize dynamic light manipulation. When the ambient sound is live music, the first auxiliary sensor **17**A assists the microprocessor **14** in synchronizing the patterns of the light with the live music.

[0047] The second auxiliary sensor **17**B is located on the interior of the base **11** and is electrically connected to the microprocessor **14**. The microprocessor **14** controls the operation of the second auxiliary sensor **17**B, and when the second auxiliary sensor **17**B generates second auxiliary sensing information according to the environment of the lighting device **10**, the microprocessor **14** receives the second auxiliary sensing information from the second auxiliary sensor **17**B and controls the light source **12** to change at least one of the color, the light intensity or the operation frequency of the light according to the second auxiliary sensing information. For example, the second auxiliary sensor **17**B is an ambient light sensor, and the second auxiliary sensing information includes the ambient light intensity of ambient light. The microprocessor **14** controls the light source **12** to change the light intensity of the light according to the ambient light intensity of the ambient light.

[0048] The power supply unit **18**A is disposed within the base **11** and is electrically connected to the light source **12**, the first wireless transceiver **13**, the microprocessor **14**, the auxiliary light sources **16**A and **16**B, the first auxiliary sensor **17**A, the second auxiliary sensor **17**B, and the rechargeable battery **18**B. It should be noted that the connection between the power supply unit **18**A and the light source **12**, the connection between the power supply unit

18A and the first wireless transceiver **13**, the connection between the power supply unit **18**A and the auxiliary light source **16**A, the connection between the power supply unit **18**A and the auxiliary light source **16**B, the connection between the power supply unit **18**A and the first auxiliary sensor **17**A, and the connection between the power supply unit **18**A and the second auxiliary sensor **17**B are omitted in order to simplify FIG. **2**B. The power supply unit **18**A receives electricity from the rechargeable battery **18**B and distributes the electricity to the light source **12**, the first wireless transceiver **13**, the microprocessor **14**, the auxiliary light sources **16**A and **16**B, the first auxiliary sensor **17**A, the second auxiliary sensor **17**B. The power supply unit **18**A may be a power management integrated circuit (PMIC), for example.

[0049] The rechargeable battery **18**B is disposed within the base **11** and is electrically connected to the battery charger **18**C to receive charging power. The rechargeable battery **18**B generates electricity to power the light source **12**, the first wireless transceiver **13**, the microprocessor **14**, the auxiliary light sources **16**A and **16**B, the first auxiliary sensor **17**A, the second auxiliary sensor **17**B, thus ensuring the sustainable operation of the lighting device **10**. For example, the rechargeable battery **18**B may be a Li-based battery, Li-ion battery, a NiMH battery, or any other suitable chemistry battery.

[0050] The battery charger **18**C is disposed within the base **11** and is electrically connected to the power transfer interface **18**D. The battery charger **18**C receives external power from the power transfer interface **18**D and generates the charging power according to the external power in order to charge the rechargeable battery **18**B.

[0051] The power transfer interface **18**D is disposed on the base **11** and is configured to receive the external power. In one embodiment, the power transfer interface **18**D is a wire power transfer interface, and the wire power transfer interface is a USB interface or a hot-plug interface, for example. The USB interface is connected to the USB cable to receive the external power from a USB socket. In another embodiment, the power transfer interface **18**D is a wireless power transfer interface, the wireless power transfer interface receives the external power from a portable charger by the electromagnetic induction of magnetic coils. In yet embodiment, the power transfer interface **18**D is provided in two, one power transfer interface **18**D is the wire power transfer interface, and the other power transfer interface **18**D is the wireless power transfer interface. Two power transfer interface **18**D may utilize the wired connection or the wireless connection to energize the lighting device **10** during operation or charging.

[0052] Please refer to FIG. **2**C, which depicts the block diagram of the electronic device shown in FIG. **2**A. As shown in FIG. **2**C, the electronic device **20** includes a processor **21**, a second wireless transceiver **22**, a memory **23** and a plurality of sensors. In the present embodiment, the number of the plurality of sensors is three, and three sensors are a first sensor **24**A, a second sensor **24**B and a third sensor **24**C. However, the number of the plurality of sensors may be adjusted according to the requirements of the electronic device **20** and may be two, four or more than four, and the present disclosure is not limited thereto. The following would elaborate the configuration of the electronic device **20** in conjunction with FIG. **2**A to FIG. **2**C.

[0053] The processor **21** is electrically connected to the second wireless transceiver **22**, the memory **23**, the first sensor **24**A, the second sensor **24**B and the third sensor **24**C, and the second wireless transceiver **22**, the first sensor **24**A, the second sensor **24**B and the third sensor **24**C are controlled by the processor **21**. The memory **22** stores user information and/or the user music, and the user information includes user music libraries, user broadcasts, user schedule, user habits, etc. The memory may be a flash memory, a hard disk drive (HDD), solid-state disk (SSD), a dynamic random access memory (DRAM), or a static random access memory (SRAM). The processor **21** is able to access the user information from the memory **22** and integrates the audio information, the sensing information, the user information and the environment information into a packet.

[0054] The second wireless transceiver **23** is wirelessly connected to the first wireless transceiver **13** (referring to FIG. **2**B) to realize the communication between the electronic device **20** and the lighting device **10** (referring to FIG. **2**A). In addition, the second wireless transceiver **23** is wirelessly connected to the cloud server **30** and the media input source **40** to receive the one part of the audio information and the environment information. The processor **21** controls the second wireless transceiver **23** to transmit the packet to the first wireless transceiver **13**. In other words, the second wireless transceiver **23** serves as the IoT interface of the electronic device **20**. The communication protocol of the second wireless transceiver **23** may be but not be limited to Bluetooth or Wi-Fi.

[0055] The first sensor **24**A generates a first sensing signal, and the processor **21** controls the second wireless transceiver **23** to transmit the first sensing signal to the first wireless transceiver **13** (referring to FIG. **2**B), and the microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change the color of the light, the light intensity of the light or the operation frequency of the light according to the first sensing signal. It should be noted that the first sensing signal is one part of the sensing information. For example, the first sensor **24**A is a velocity sensor, and the first sensing signal represents velocity information. The microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change the operation frequency of the light and/or the color of the light according to the velocity information.

[0056] The second sensor **24**B generates a second sensing signal. It should be noted that the second sensing signal is another part of the sensing information. For example, the second sensor **24**B is a GPS sensor, and the second sensing signal represents location information. The location information includes the geographical orientation of the electronic device **20** and the geographical position of the electronic device **20**. The processor **21** selects the local music according to the geographical orientation of the electronic device **20** and the geographical position of the electronic device **20** and transmits the local music to the lighting device **10** by the second wireless transceiver **23**. The microprocessor controls the light source **12** and the auxiliary light sources **16**A and **16**B to change at least one of the colors, the light intensity and the operation frequency of the light according to the local music.

[0057] The third sensor **24**C generates a third sensing signal, and the processor **21** controls the second wireless transceiver **23** to transmit the third sensing signal to the first wireless transceiver **13** (referring to FIG. **2**B), and the microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change the color of the light, the light intensity of the light or the operation frequency of the light according to the third sensing signal. It should be noted that the third sensing signal is the other part of the sensing information. For example, the third sensor **24**C is an accelerometer, and the third sensing signal represents acceleration information. The microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change the operation frequency of the light and/or the color of the light according to the acceleration information.

[0058] In one embodiment, the processor **21** obtains the user information from the memory **22**, selects the user music from the audio information according to the user information, and controls the second wireless transceiver **23** to transmit the user music to the first wireless transceiver **13** (referring to FIG. **2**B). The microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change at least one of the colors of the light, the light intensity of the light, and the operation frequency of the light according to the user music.

[0059] In another embodiment, the processor **21** obtains the user music from the memory **22** and controls the second wireless transceiver **23** to transmit the user music to the first wireless transceiver **13** (referring to FIG. **2**B). The microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change at least one of the colors of the light, the light intensity of the light, and the operation frequency of the light according to the user music.

[0060] In the present embodiment, there are two auxiliary sensors including the microphone and the ambient light sensor, and there are three sensors including the velocity sensor, the GPS sensor, and the accelerometer. In another embodiment, there are no auxiliary sensors, and there are five sensors including the microphone, the ambient light sensor, the velocity sensor, the GPS sensor, and the accelerometer. The sensing signal corresponding to the microphone represents the frequency and the sound level of ambient sound around the environment of the electronic device **20**. The sensing signal corresponding to the ambient light sensor represents the ambient light intensity of the ambient light.

[0061] Please refer to FIG. **3**, which depicts a flowchart of a method of lighting in a beverage container according to one embodiment. As shown in FIG. **3**, the method of lighting in the beverage container includes step **S11** and step **S12**. The method of lighting in the beverage container may be applicable to the lighting system shown in FIG. **1**A and the lighting device **10** shown in FIG. **1**B but not be limited thereto. Step **S11** and step **S12** would be exemplarily explained by the lighting system shown in FIG. **1**A and the lighting device **10** shown in FIG. **1**B as follows.

[0062] Step **S11**: receiving the audio information and the sensing information. As described above, the lighting device **10** receives the audio information and the sensing information from the electronic device **20**. The audio information includes user music, popular music, local music and the other audio files. The sensing information includes a wealth of sensing signals, and the sensing signals, for example, represent the frequency and the sound level of the ambient sound around the environment of the electronic device **20**, the ambient light intensity of the ambient light, the velocity information, the location information, and the acceleration information.

[0063] Step **S12**: according to the audio information and sensing information, changing at least one of the colors, the light intensity, and the operation frequency of the light.

[0064] With regard to the frequency and the sound level of the ambient sound around the environment of the electronic device **20**, the microprocessor **14** maps the frequency of the ambient sound around the environment of the electronic device **20** into a light wavelength by a lookup table or an algorithm. For example, the frequency of the ambient sound within the range of 0.5 kHz to 25 kHz would be transformed into the light wavelength within the range of 400 nm to 700 nm. The microprocessor **14** maps the sound level of the ambient sound around the environment of the electronic device **20** into the light intensity by the lookup table or the algorithm. Afterwards, the microprocessor **14** controls the light source **12** to change the color and the light intensity of the light based on the mapped light intensity and the mapped light wavelength. In other words, the variations of the ambient sound around the environment are mapped into the variations of the pattern of the light to create a wide light spectrum illumination that couples in harmony with the ambient sound and to allow for the captivating visual representation of the ambient sound.

[0065] With regard to the ambient light intensity of the ambient light, the microprocessor **14** controls the light source **12** to adjust the light intensity of the light according to the ambient light intensity of the ambient light. When determining that the ambient light intensity is greater than a light intensity threshold, the microprocessor **14** controls the light source **12** to decrease the light intensity of the light; when determining that the ambient light intensity is not greater than the light intensity threshold, the microprocessor **14** controls the light source **12** to increase the light intensity of the light.

[0066] With regard to the location information, the electronic device **20** selects the local culture and the ambiance where the electronic device **20** is located from the geographical information according to the geographical orientation and the geographical position of the electronic device **20** in the location information and selects the local music reflecting the local culture and the ambiance at the geographical orientation of the electronic device **20** accordingly. Afterwards, the lighting device **10** receives the local music from the electronic device **20** by the first wireless transceiver **13**, and the microprocessor **14** controls the light source **12** to change the color and the light intensity of the light based on the local music.

[0067] With regard to the velocity information and the acceleration information, the electronic device **20** determines a user action according to the velocity information, the acceleration information, and the geographical orientation of the electronic device **20**, and the lighting device **10** receives the user action from the electronic device **20** by the first wireless transceiver **13**. Afterwards, the microprocessor **14** controls the light source **12** to change the operation frequency of the light according to the user action. When the user action is a static action, the microprocessor **14** controls the light source **12** to emit static light. When the user action is a dynamic action such as shaking, the microprocessor **14** controls the light source **12** to emit dynamic light and synchronizes the operation frequency of the light with the velocity of the user, and the dynamic light can strobe and pulse in rhythm with the user action, thus intensifying the user experience of the beverage container with the lighting device when the user begins to dance or shake.

[0068] In another embodiment, the electronic device **20** selects the user music or the needed light template according to the user information, and the lighting device **10** receives the user music or the needed light template from the electronic device **20** by the first wireless transceiver **13**. The microprocessor **14** controls the light source **12** to change the color of the light and the light intensity of the light according to the user music, or controls the light source **12** to emit the light by the needed light template to provide an enriched audio experience tailored to the user preferences.

[0069] In yet embodiment, the electronic device **20** transmits the weather information and the time information to the lighting device **10**. The microprocessor **14** controls the light source **12** to change the color of the light and the light intensity of the light according to the weather information and the time information.

[0070] In the method of lighting in the beverage container of the present embodiment, the audio information and the sensing information are mapped into the variations of the pattern of the light encompassing the adjustments of the color and the light intensity of the light and the strobe effects to enhance the visual experience of the user and to create the dynamic interplay between the light and the sound.

[0071] Please refer to FIG. **4**, which depicts a flowchart of a method of lighting in a beverage container according to another embodiment. As shown in FIG. **4**, the method of lighting in the beverage container includes step S**21** to step S**24**, and step S**21** and step S**24** are similar to step S**11** and step S**12** and would not be repeated. Step S**22** and step S**23** would be exemplarily explained by the lighting system shown in FIG. **2**A, the lighting device **10** shown in FIG. **2**B and the electronic device **20** shown in FIG. **2**C as follows.

[0072] Step S**22**: generating the auxiliary sensing information according to the environment of the lighting device **10**. As described above, the first auxiliary sensor **17**A generates the first auxiliary sensing information according to the environment of the lighting device **10**, and the second auxiliary sensor **17**B generates second auxiliary sensing information according to the environment of the lighting device **10**. For example, the first auxiliary sensing information includes the frequency and the SPL of the ambient sound around the environment of the lighting device **10**, and the second auxiliary sensing information includes the ambient light intensity of the ambient light.

[0073] Step S**23**: changing at least one of the colors, the light intensity and the operation frequency of the light according to the auxiliary sensing information.

[0074] With regard to the frequency and the SPL of the ambient sound around the environment of the lighting device **10**, the microprocessor **14** maps the frequency of the ambient sound around the environment of the lighting device **10** into the light wavelength by the lookup table or the algorithm and maps the SPL of the ambient sound around the environment of the lighting device **10** into the light intensity by the lookup table or the algorithm. Afterwards, the microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change the color and the light intensity of the light based on the mapped light intensity and the mapped light wavelength.

[0075] With regard to the ambient light intensity of the ambient light, the microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to adjust the light intensity of the light according to the ambient light intensity of the ambient light.

[0076] In the method of lighting in the beverage container of the present embodiment, the auxiliary sensing information assists the lighting device **10** in fostering a deep emotional band with the user through the interactive engagement of the lighting effects.

[0077] Please refer to FIG. **5**, which depicts a flowchart of a method of lighting in a beverage container according to yet embodiment. As shown in FIG. **5**, the method of lighting in the beverage container includes step S**31** to step S**36**. Step S**31** to step S**36** would be exemplarily explained by the lighting system shown in FIG. **2**A, the lighting device **10** shown in FIG. **2**B and the electronic device **20** shown in FIG. **2**C.

[0078] Step S**31**: integrating the lighting device **10** with the beverage container C**1**. Specifically, the lighting device **10** is attached to the bottom surface of the beverage container C**1** by the attaching surface A**1** to embed the lighting device **10** in the beverage container C**1**.

[0079] Step S**32**: paired to the electronic device **20**. Specifically, the user utilizes the application software to give a pairing instruction to the processor **21**, and the processor **21** pairs the electronic device **20** and the lighting device **10**.

[0080] Step S**33**: controlling the operation of the lighting device **10**. When the electronic device **20** is paired with the lighting device **10**, the user utilizes the application software to control the processor **21** to generate control commands, and the microprocessor **14** receives the control commands from the electronic device **20** by the first wireless transceiver **13**. Afterwards, the microprocessor **14** operates according to the control commands. In other words, the user utilizes the application software to indirectly control the operation of the lighting device **10**.

[0081] Step S**34**: being connected to the cloud server **30**. Specifically, the user utilizes the application software to give a connection instruction to the processor **21**, and the processor **21** is connected to the cloud server **30** by the second wireless transceiver **23** according to the connection instruction. In addition, the processor **21** can be connected to the media input source **40** and the external storage device **50** by the second wireless transceiver **23**. The processor **21** gathers the sensing information and the audio information from the cloud server **30**, the media input source **40**, and the external storage device **50**.

[0082] Step S**35**: requesting customer details. In the present embodiment, the application software shows a prompting message to the user, and the user selects the needed light template in response to the prompting message. Afterwards, the microprocessor **14** receives the needed light template by the first wireless transceiver **13**.

[0083] In another embodiment, the application software shows a prompting message to the user, and the user selects the user music in response to the prompting message. Afterwards, the microprocessor **14** receives the user music by the first wireless transceiver **13**.

[0084] Step S**36**: personalizing the user experience. In the present embodiment, the microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change the color, the light intensity and the operation frequency of the light to personalize the lighting effects of the lighting device **10** and to enhance the user experience of the beverage container with the lighting device.

[0085] In another embodiment, the microprocessor **14** controls the light source **12** and the auxiliary light sources **16**A and **16**B to change the color, the light intensity and the operation frequency of the light to customize the lighting effects of the lighting device **10**.

[0086] In view of the above description, in the lighting device for the beverage container, the lighting device and the beverage container integrate seamlessly, and the pattern of the light varies within the beverage container according to the audio information and the sensing information, thereby creating vibrant visual experience, thus enhancing the user experience of the beverage container with the lighting device and encouraging interactive communication with customers.

[0087] In view of the above description, in the lighting system for the beverage container and the method of lighting in the beverage container, the lighting device receives plenty of real-time data such as the audio information and the sensing information from the electronic device to synchronize the pattern of the light with the music in the audio information and to transform the sensing information into the dynamic pattern of the light, thereby fostering an enriching experience with the beverage container having the lighting device.

[0088] In addition, when the beverage container with the lighting device is exhibited, the beverage container with the lighting device not only engage consumers at a deeper level but also transforms the consumer behavior of purchasing and experiencing beverage in the beverage container into a more immersive and enjoyable journey.

What is claimed is:

1. A lighting device for a beverage container comprising:

a base affixed to the beverage container;

at least one light source disposed on the base and configured to emit light;

a wireless transceiver disposed within the base and configured to receive audio information and sensing information; and

a microprocessor disposed within the base and electrically connected to the at least one light source and the wireless transceiver, wherein the microprocessor controls the at least one light source to change at least one of colors, light intensity, and an operation frequency of the light according to the audio information and the sensing information.

2. The lighting device for the beverage container according to claim **1**, further comprising:

at least one auxiliary sensor disposed on the base, electrically connected to the microprocessor, and generating and transmitting auxiliary sensing information to the microprocessor according to an environment of the lighting device, wherein the microprocessor controls the at least one light source to change at least one of the colors, the light intensity or the operation frequency of the light according to the auxiliary sensing information.

3. The lighting device for the beverage container according to claim **1**, wherein the lighting device is connected to an electronic device by the wireless transceiver to receive the sensing information and obtains environment information and the audio information from a cloud server, a media input source, and an external storage device by the electronic device, and the microprocessor adjusts at least one of the colors, the light intensity, and the operation frequency of the light according to the environment information.

4. The lighting device for the beverage container according to claim **3**, wherein the electronic device selects user music from the audio information according to user information and transmits the user music to the lighting device, and the step of controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information by the microprocessor comprises:

according to the user music, controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light.

5. The lighting device for the beverage container according to claim **3**, wherein the sensing information comprises location information, the electronic device selects local music from the audio information according to location information and transmits the local music to the lighting device, and the step of controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information by the microprocessor comprises:

according to the local music, controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light.

6. The lighting device for the beverage container according to claim **1**, wherein the sensing information comprises a frequency and a sound level of ambient sound and velocity information, and the step of controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information by the microprocessor comprises:

according to the frequency of the ambient sound, controlling the at least one light source to change the colors of the light;

according to the sound level of the ambient sound, controlling the at least one light source to change the light intensity of the light; and

according to the velocity information, controlling the at least one light source to change the operation frequency of the light.

7. The lighting device for the beverage container according to claim **1**, further comprising:

a switch disposed adjacent to the base, electrically connected to the microprocessor and configured to generate a trigger signal, wherein the microprocessor controls the at least one light source to be turned on or be turned off according to the trigger signal;

a rechargeable battery disposed within the base and generating electricity;

a power supply unit disposed within the base, electrically connected to the rechargeable battery, the at least one light source, the wireless transceiver, and the microprocessor, and distributing the electricity to the at least one light source, the wireless transceiver and the microprocessor;

a power transfer interface disposed on the base and configured to receive external power by wired power transfer or wireless power transfer; and

a battery charger disposed within the base, electrically connected to the rechargeable battery and the power transfer interface, and charging the rechargeable battery according to the external power.

8. The lighting device for the beverage container according to claim **1**, wherein the base comprises an attaching surface to be attached to the container.

9. The lighting device for the beverage container according to claim **1**, wherein the lighting device is wirelessly connected to a phone application, and the phone application collects user information, the sensing information and the audio information from a cloud server, a media input source and an external storage device and remotely controls operation of the lighting device.

10. A lighting system for a beverage container comprising:

an electronic device transmitting audio information and sensing information;

a lighting device communicating with the electronic device and comprising:

a base affixed to the beverage container;

at least one light source disposed on the base and configured to emit light;

a wireless transceiver disposed within the base and configured to receive the audio information and the sensing information;

a microprocessor disposed within the base and electrically connected to the at least one light source and the wireless transceiver, wherein the microprocessor controls the at least one light source to change at least one of colors, light intensity, and an operation frequency of the light according to the audio information and the sensing information.

11. The lighting system for the beverage container according to claim **10**, wherein the lighting device further comprising:

at least one auxiliary sensor disposed on the base, electrically connected to the microprocessor, and generating and transmitting auxiliary sensing information to the microprocessor according to an environment of the lighting device, wherein the microprocessor controls the at least one light source to change at least one of the colors, the light intensity or the operation frequency of the light according to the auxiliary sensing information.

12. The lighting system for the beverage container according to claim **10**, wherein the electronic device comprises a plurality of sensors configured to generate the sensing information and obtains environment information and the audio information from a cloud server, a media input source and an external storage device; the lighting device receives the environment information from the electronic device, and the microprocessor adjusts at least one of the colors, the light intensity, and the operation frequency of the light according to the environment information.

13. The lighting system for the beverage container according to claim **12**, wherein the plurality of sensors comprises a microphone and a velocity sensor, and the sensing information comprises a frequency and a sound level of ambient sound and velocity information; the step of controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information by the microprocessor comprises:

according to the frequency of the ambient sound, controlling the at least one light source to change the colors of the light;

according to the sound level of the ambient sound, controlling the at least one light source to change the light intensity of the light; and

according to the velocity information, controlling the at least one light source to change the operation frequency of the light.

14. The lighting system for the beverage container according to claim **12**, wherein the plurality of sensors comprise a GPS sensor, the sensing information comprises location information, and the processor selects local music from the audio information according to location information; the lighting device receives the local music from the electronic device, and the step of controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information by the microprocessor comprises:

according to the local music, controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light.

15. The lighting system for the beverage container according to claim **10**, wherein the electronic device comprises a processor and a memory electrically connected to the processor, and the processor obtains user information from the memory and selects user music from the audio information according to the user information; the lighting device receives the user music from the electronic device, and the step of controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information by the microprocessor comprises:

according to the user music, controlling the at least one light source to change at least one of the colors, the light intensity, and the operation frequency of the light.

16. The lighting system for the beverage container according to claim **10**, wherein the electronic device is provided with application software, and the application software gathers user information, the sensing information and the audio information and controls lighting of the lighting device.

17. A method of lighting in a beverage container, for a lighting system comprising an electronic device and a lighting device configured to emit light and affixed to the beverage container, comprising:

performing steps as follows by the lighting device:

receiving audio information and sensing information from the electronic device; and

according to the audio information and sensing information, changing at least one of colors, light intensity, and an operation frequency of the light.

18. The method of lighting in the beverage container according to claim **17**, further comprising steps performed by the lighting device:

generating auxiliary sensing information according to an environment of the lighting device; and

according to the auxiliary sensing information, changing at least one of the colors, the light intensity, and the operation frequency of the light.

19. The method of lighting in the beverage container according to claim **17**, further comprising:

performing steps as follows by the electronic device:

generating the sensing information, and obtaining environment information and the audio information from a cloud server, a media input source, and an external storage device;

performing steps as follows by the lighting device:

receiving the environment information from the electronic device; and

according to the environment information, changing at least one of the colors, the light intensity, and the operation frequency of the light.

20. The method of lighting in the beverage container according to claim **17**, wherein the sensing information comprises a frequency and a sound level of ambient sound and velocity information, and changing at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information comprising:

according to the frequency of the ambient sound, changing the colors of the light;

according to the sound level of the ambient sound, changing the light intensity of the light;

according to the velocity information, changing the operation frequency of the light.

21. The method of lighting in the beverage container according to claim **17**, further comprising:

performing steps as follows by the electronic device:

obtaining user information, and selecting user music from the audio information according to the user information;

performing steps as follows by the lighting device:

receiving the user music from the electronic device;

changing at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information comprising:

according to the user music, changing at least one of the colors, the light intensity, and the operation frequency of the light.

22. The method of lighting in the beverage container according to claim **17**, wherein the sensing information comprises location information, further comprising:

performing steps as follows by the electronic device:

selecting local music from the audio information according to location information;

performing steps as follows by the lighting device:

receiving the local music from the electronic device;

changing at least one of the colors, the light intensity, and the operation frequency of the light according to the audio information and sensing information comprising:

according to the local music, changing at least one of the colors, the light intensity, and the operation frequency of the light.

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